THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Mass is conserved in a: 1

(a) nuclear change only **(b)** chemical reaction
(c) no reaction **(d)** only physical change

2. A positive ion is called: 1

(a) atom **(b)** anion **(c)** molecule **(d)** cation

3. The bond in NaCl is: 1

(a) hydrogen **(b)** metallic only **(c)** covalent **(d)** ionic

4. The law of conservation of mass was given by: 1

(a) Newton **(b)** Dalton **(c)** Lavoisier **(d)** Proust

5. A negative ion is called: 1

(a) atom **(b)** cation **(c)** molecule **(d)** anion

6. The bond in $\mathrm{H_2}$ is: 1

(a) covalent **(b)** ionic **(c)** hydrogen bond **(d)** metallic

7. Water contains hydrogen and oxygen in the ratio: 1

(a) 8 : 1 **(b)** 1 : 1 **(c)** 2 : 1 **(d)** 1 : 8

8. Sodium ion is: 1

(a) $\mathrm{Na^{2+}}$ **(b)** $\mathrm{Na^{+}}$ **(c)** $\mathrm{Na^{2-}}$ **(d)** $\mathrm{Na^{-}}$

9. Ionic compounds are usually: 1

(a) low melting liquids **(b)** soft solids
(c) gases **(d)** solids with high melting point

10. Atoms are the building blocks of: 1

(a) matter **(b)** only gases **(c)** energy **(d)** only metals

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the law of conservation of mass. 2

12. What is an ion? 2

13. Name the bond formed by transfer of electrons. 2

14. State the law of constant proportions. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Write three postulates of Dalton's atomic theory. 3

16. List three properties of covalent compounds. 3

17. How do atoms of a non-metal form ions? Give two examples. 3

OR

Find the mass of CO2 formed when 12 g of carbon burns in 32 g of oxygen and name the law used. 3

18. Write the atomicity of: He, $\mathrm{O_3}$, $\mathrm{P_4}$, $\mathrm{S_8}$, $\mathrm{H_2}$, $\mathrm{CH_4}$. 3

OR

Write the symbols of the ions: sodium, calcium, aluminium. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. State and explain the law of conservation of mass with an example. 5

OR

Write the formulae of five compounds using valency or charges. 5

20. State and explain the law of constant proportions with an example. 5

OR

Calculate the molecular masses of five compounds. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND